REP-035 · More fruit on the farmer cards, and computers that use specialities

Game	Fruit Market (rules version 4)
Report date	2026-10-27 16:42 (Asia/Kuala_Lumpur)
Type	Balance change and computer-player change, with same-seed comparisons
Plan	PLAN-017, approved: "approve option 1 and 2" (REP-034 section 5)
Branch	`workflow/plan-017` , merged to `main` for the preview. **Live game unchanged**

Labels. [Sim] = computer players, 300 games per test, the same seeds as REP-033/034, about ±5 points. **[Hyp]** = a guess about people.

1. Changes

Option 2: more fruit (rule change). Every fruit listed on a farmer card brings 1 more. Printed values are now 3–5 (were 2–4). The unlisted fruit still comes as 1; the 2-shop and 4-shop adjustments are unchanged. `RULES.md` updated.

Option 1: smarter computers (no rule change). Normal and Hard computers now:

- with a **Juice Bar**, count the $1 tip on older fruit and the $2 juice floor, so cheap old fruit is worth buying;
- with a **Cold Room**, value stock as lasting about twice as long, and hold back on the nights their fruit doesn't age;
- stock up harder the day before a **Market Day**;
- pick the speciality that suits them (old fruit about → Juice Bar; Market Days to come → Cold Room), so plans vary across the table.

Easy computers are unchanged.

2. Results [Sim]

Measure (3 players unless noted)	Before (REP-034)	More fruit only	Both changes
Hard beats Normal (2 players)	56%	60%	**60%**
Hard beats Easy (2 players)	66%	75%	**82%**
Normal beats Easy (2 players)	56%	—	**72%**
Comeback from last at half-time	28%	37%	**38%**
Day-5 leader wins	32%	35%	31%
Blowouts (2 / 3 / 4 players)	36 / 27 / 14%	38 / 23 / 18%	42 / 23 / 15%
Big Basket trait (fair 33%)	40%	48%	**45%**
Other traits	31–34%	26–35%	27–33%

Measure (3 players unless noted)	Before (REP-034)	More fruit only	Both changes
Skip Cold Room / skip Juice Bar	32% / 35%	36% / 32%	33% / 34%
Skip Barrow	21%	21%	**16%**
Juice Bar built per player (2 players)	0.06	0.30	0.35
Sales within $1 of the top price	90%	86%	86%
Average final cash (2 players, 10 days)	$125	$162	$162

3. What this means

Good.

- Skill now matters clearly: Hard beats Normal 60% (the earlier aim was 62–65%), and Easy is a real beginner level.
- Comebacks rose to 38%, and day-5 leaders win less often. Games stay open longer.
- Customers are served more often (46% of wanted fruit, was 31%).

Not fixed: Cold Room and Juice Bar.

- The REP-033 "rush" test was misleading. A shop that rushes one speciality also skips the Barrow, and skipping the Barrow alone drops a shop to 16%.
- The fairer measure is "skip it": skipping Cold Room or Juice Bar still makes no difference (33–34%). These two are harmless but add little. Even with the smarter computers, almost no fruit spoils, so the Juice Bar has little to do.

New problem: volume wins.

- With more fruit, buying more each day is the strongest plan: Big Basket wins 45% and the Barrow is close to essential.
- 2-player blowouts rose to 42%.

4. Validation

Check	Result	Evidence
Same-seed comparisons (section 2)	PASS (run and recorded)	3 runs of the balance report, 300 games per test
`npm run verify`	PASS	Local run before merge
CI `verify:full` on `main`	Reported after the push	GitHub Actions CI
PLAN-016 CI on `main`	PASS	CI run 26 green, preview updated

5. Options (need your decision; nothing changed yet)

1. **Rein in volume.** Big Basket becomes "buy 1 extra fruit on Market Days", or the Barrow becomes "hold 12" only (no extra buy). Recommended: test both in the lab and bring you the numbers first.

2. **Give the Juice Bar and Cold Room a sharper job.** For example, Juice Bar: "buy not-so fruit at the market for $0".
3. **Playtest first** and see whether people find the same.